OAS-002 — Architectural Patterns

Status: Foundational

Authority: Architectural

Depends on:

- OAS-000
- OAS-001
- OCS-000 through OCS-032

Purpose

This specification defines the architectural patterns of Organodynamics.

Architectural Patterns are reusable organizations of constitutional behavior.

A Pattern is not a component.

A Pattern is not an implementation.

A Pattern is a proven architectural arrangement that preserves constitutional principles while solving recurring architectural problems.

Patterns SHALL remain implementation-independent.

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1. Definition

An Architectural Pattern is a repeatable constitutional organization of responsibilities, relationships, and interactions.

Every Pattern SHALL expose:

Purpose

Context

Responsibilities

Boundaries

Dependencies

Expected Behavior

Known Tradeoffs

Patterns SHALL remain independently understandable.

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2. Purpose

Architectural Patterns exist to preserve constitutional integrity while allowing architectural diversity.

Patterns SHALL:

Reduce unnecessary complexity.

Improve architectural consistency.

Preserve replaceability.

Support independent implementation.

Patterns SHALL never reduce constitutional freedom.

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3. Pattern Identity

Every Architectural Pattern SHALL possess:

Stable Identity

Canonical Name

Architectural Purpose

Historical Lineage

Revision History

Patterns SHALL evolve without losing identity.

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4. Pattern Scope

Every Pattern SHALL declare:

Applicable Context

Architectural Layer

Expected Constraints

Known Limitations

Non-applicable Contexts

Patterns SHALL NOT silently expand beyond their declared scope.

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5. Pattern Composition

Patterns MAY compose with other Patterns.

Composition SHALL preserve:

Local responsibilities.

Architectural boundaries.

Constitutional behavior.

Independent evolution.

Composition SHALL remain explainable.

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6. Pattern Independence

Patterns SHALL remain independent of:

Programming languages.

Databases.

Frameworks.

Cloud providers.

AI models.

Deployment strategies.

Technology SHALL instantiate Patterns.

It SHALL not define them.

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7. Pattern Selection

Architects MAY select different Patterns to satisfy equivalent constitutional requirements.

Selection SHALL remain explainable.

Equivalent constitutional behavior SHALL remain preserved.

Architectural diversity SHALL be encouraged.

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8. Pattern Evolution

Patterns MAY evolve through architectural experience.

Evolution SHALL preserve:

Identity.

Purpose.

Historical lineage.

Constitutional compatibility.

Patterns SHALL evolve more slowly than implementations.

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9. Pattern Relationships

Patterns MAY:

Extend other Patterns.

Refine other Patterns.

Compose other Patterns.

Specialize other Patterns.

Relationships SHALL remain explicit.

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10. Pattern Verification

Every Pattern SHALL define:

Expected architectural properties.

Expected constitutional preservation.

Known failure modes.

Verification criteria.

Architectural verification SHALL remain implementation-independent.

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11. Pattern Catalog

The Organodynamics Architectural Specification SHALL maintain a Pattern Catalog.

The Catalog SHALL preserve:

Canonical definitions.

Pattern relationships.

Historical evolution.

Known applications.

Patterns SHALL remain independently discoverable.

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12. Pattern Reuse

Architectural reuse SHALL occur through Patterns.

Reuse SHALL preserve:

Meaning.

Responsibilities.

Architectural boundaries.

Historical continuity.

Copying implementation SHALL NOT constitute architectural reuse.

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13. Pattern Failure

Architectural Pattern failure SHALL occur when:

Responsibilities become ambiguous.

Boundaries disappear.

Constitutional behavior changes.

Replaceability is reduced.

Historical reasoning is lost.

Failure SHALL become architectural knowledge.

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14. Architectural Pattern Test

An Architectural Pattern satisfies this specification when independent architects, applying the Pattern in different implementation environments, produce constitutionally equivalent architectural behavior while retaining architectural diversity.

Conformance

An architecture conforms to this specification when every reusable architectural organization is explicitly represented as an Architectural Pattern whose constitutional purpose, responsibilities, and boundaries remain reconstructable independently of implementation technology.

Architectural Principle

Patterns are the memory of architecture.

They preserve not finished designs,

but successful ways of organizing meaning.

Every generation of architects should therefore inherit Patterns,

not prescriptions,

so that architecture may evolve without forgetting why earlier solutions worked.

End of Specification OAS-002